



What's LLM? Understanding Large Language Models

Dive into the fascinating world of Large Language Models and discover how AI understands and generates human language with remarkable precision.

What is an LLM?

Large Language Models (LLMs) are sophisticated AI systems trained on massive datasets containing billions of words from books, articles, and web content. These powerful models learn to understand and generate human-like language with remarkable fluency and accuracy.

Today, LLMs power intelligent chatbots, real-time translators, content creation tools, and countless other applications that transform how we interact with technology.



Popular LLMs:

Open Source LLM	Close Sourced LLM
OpenAI's GPT-OSS,	OpenAI's ChatGPT,
Google Gemma	Anthropic Claude,
Meta's LLaMA,	Google's Gemini,
Perplexity Sonar	
Grok	
DeepSeek,	
Qwen	

How Do LLMs Work? The Transformer Architecture

Neural Networks

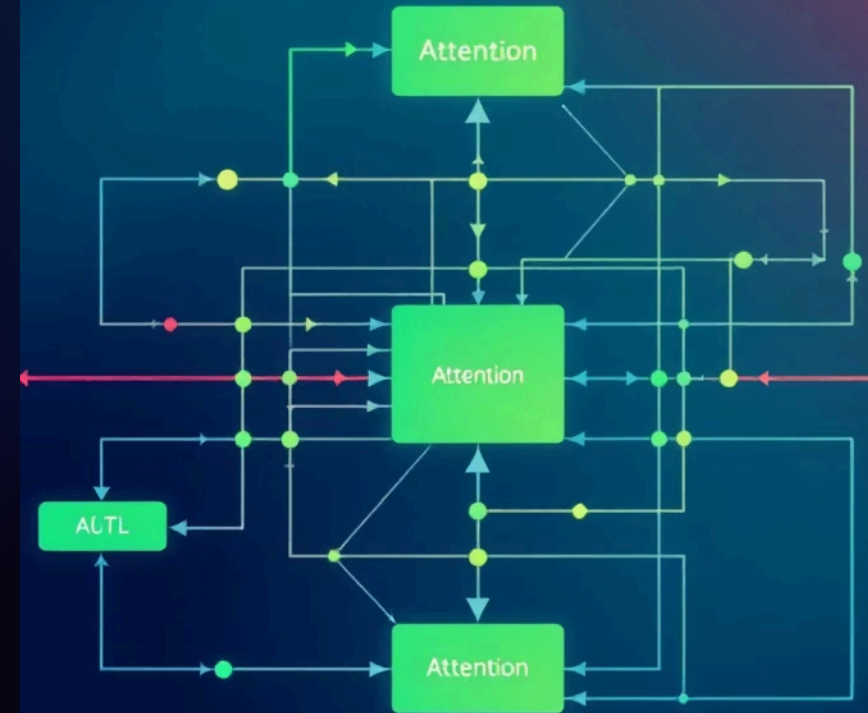
Built on advanced transformer architectures with billions of interconnected parameters that process language patterns

Attention Mechanisms

Sophisticated systems that focus on relevant parts of input text, understanding context and relationships between words

Parallel Processing

Process entire sequences simultaneously rather than word-by-word, enabling lightning-fast, context-aware language understanding



Tokens: The Building Blocks of Language Models

Text is systematically broken down into smaller units called tokens—these can be complete words, parts of words, or even individual characters. LLMs use these tokens as the fundamental building blocks for understanding and generating language.

The model predicts the next most likely token based on all previous tokens in the sequence, creating coherent and contextually appropriate text through this iterative process.



- ❗ **Example:** "The quick brown fox" gets processed as individual tokens, with the model predicting "jumps" as the most probable next word based on learned patterns.

Context

what the model knows right now from the input you provide.

- **Context window: The amount of text (tokens) the model can “see” at once.**
- **For example, if a model has a 8k token context window, it can only process about 8,000 tokens of conversation history, instructions, or documents at a time.**

Context Window: How Much Can an LLM Remember?

The context window represents the maximum number of tokens an LLM can actively consider and remember during any single interaction or task.

128K

Modern Models

Advanced LLMs like GPT-4 Turbo can handle massive context windows equivalent to hundreds of pages of text

!M

Cutting Edge

Latest models push boundaries even further, processing entire books or lengthy research papers in one go

10x

Performance Boost

Larger context windows dramatically improve understanding of long documents and maintain conversation coherence





Prompt Engineering: Guiding the Model's Output

Prompt engineering is the art and science of crafting precise inputs to elicit the most accurate, relevant, and useful responses from language models.

01

Zero-Shot Prompting

Asking the model to perform tasks without providing any examples, relying solely on its pre-trained knowledge

02

Few-Shot Prompting

Including 2-5 examples in your prompt to demonstrate the desired format or approach for the task

03

Chain-of-Thought

Encouraging the model to show its reasoning process step-by-step for complex problem-solving tasks

Hyperparameters: The Model's Settings



Hyperparameters are crucial configuration settings that determine how an LLM learns during training and behaves during inference. These parameters are set before training begins and significantly influence the model's ability to learn complex language patterns.

- **Learning Rate:** Controls how quickly the model updates its understanding
- **Batch Size:** Determines how many examples are processed simultaneously
- **Temperature:** Adjusts creativity vs. predictability in outputs
- **Layer Count:** Defines the depth of the neural network architecture

Fine-tuning these parameters allows researchers to optimise models for specific tasks or domains, dramatically improving performance on specialised applications.

Model Weights: The Knowledge Inside

175 Billion

Parameters in GPT-3, each storing learned language patterns



Neural Connections

Weights are numerical values that represent the strength of connections between artificial neurons, learned through exposure to vast amounts of text data



Encoded Knowledge

Billions of weights capture intricate patterns including grammar rules, factual information, reasoning abilities, and cultural understanding



Language Processing

These weights determine precisely how the model transforms input tokens into meaningful, contextually appropriate, and linguistically correct output



AI

Key LLM Terminologies Recap

1

Foundation Concepts

LLMs, Transformers, Tokens -

The core building blocks that enable AI to process and generate human language

2

Control Mechanisms

Context Windows, Prompts, Hyperparameters - Tools for managing and optimising model behavior and performance

3

Internal Representations

Weights, Embeddings - The mathematical foundations that store knowledge and enable semantic understanding

Understanding these fundamental concepts empowers you to work more effectively with AI language models and appreciate the remarkable engineering behind modern conversational AI systems.